EXPERIMENT NO. 8

Experiment Name: NMAP to Discover Live Hosts Using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe Platform

Aim

To discover live hosts in a network using the Nmap tool and perform different types of network scans, including ARP, ICMP, TCP and UDP scans, on the TryHackMe platform. The experiment aims to understand host discovery techniques, identify reachable systems and observe how different scanning methods determine network availability.

Procedure

1. Open a web browser and visit the TryHackMe platform. Log in to the account and open the assigned room related to Nmap and network scanning.
2. Start the target machine or lab environment provided by TryHackMe. Note the target IP address displayed by the platform. All scans must be performed only against the assigned lab target or another system for which permission has been granted.
3. Open the terminal provided by the TryHackMe environment. Verify that Nmap is installed by executing the following command:

nmap --version

This command displays the installed Nmap version and confirms that the tool is available for use.

4. **ARP Scan:** ARP (Address Resolution Protocol) is used to discover devices on the local IPv4 network by mapping IP addresses to MAC addresses. Execute the following command on the authorized target network:

sudo nmap -PR <TARGET_IP>

Replace <TARGET_IP> with the assigned target IP address. Observe the scan output to determine whether the host is reachable. ARP discovery is mainly useful for devices on the same local network segment.

5. **ICMP Scan:** ICMP (Internet Control Message Protocol) is commonly used to check whether a host is reachable. Perform the following command:

sudo nmap -PE <TARGET_IP>

Observe the output to identify whether the target responds to ICMP echo requests. A missing response does not always mean that the host is offline because firewalls may block ICMP packets.

6. **TCP Scan:** TCP-based discovery checks whether a host responds to TCP probes on selected ports. Execute:

sudo nmap -PS80 <TARGET_IP>

This command sends TCP SYN probes to port 80. Examine the output to understand whether the target responds to the probe. The result depends on the target's network configuration, firewall rules and port availability.

7. **UDP Scan:** UDP is a connectionless transport-layer protocol. Perform a UDP-based host discovery probe using:

sudo nmap -PU53 <TARGET_IP>

This command sends UDP probes to port 53. Observe the response and compare it with the results of the ARP, ICMP and TCP probes. UDP scans may produce limited results because some hosts and firewalls do not respond to UDP probes.

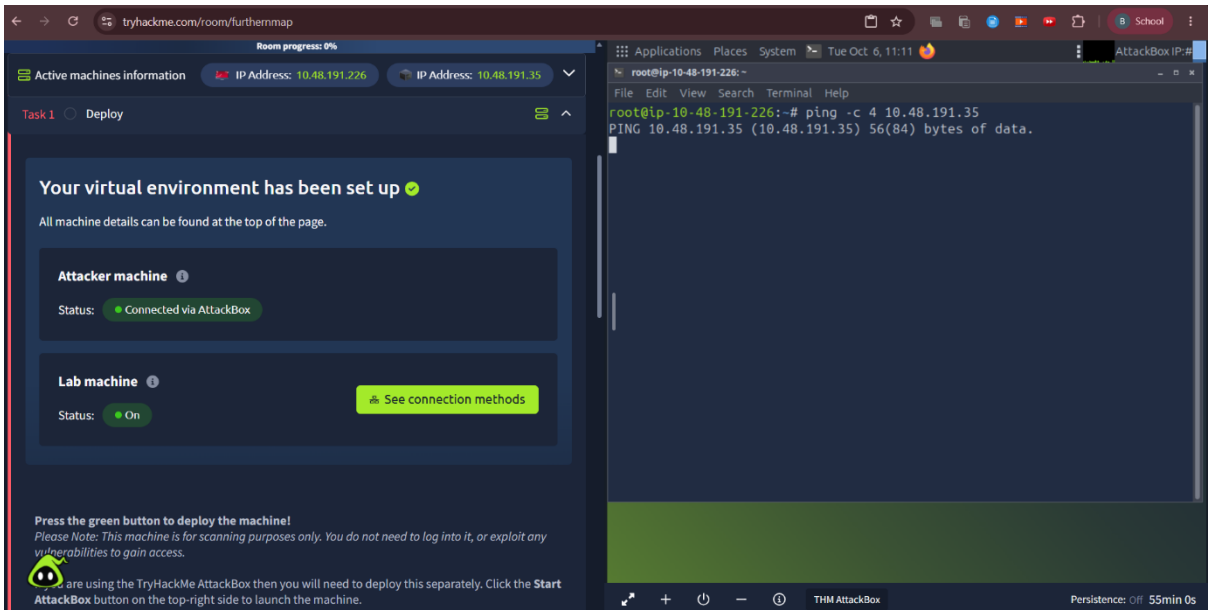
8. **General Nmap Scan:** Execute the following command to perform a basic scan of the authorized target:

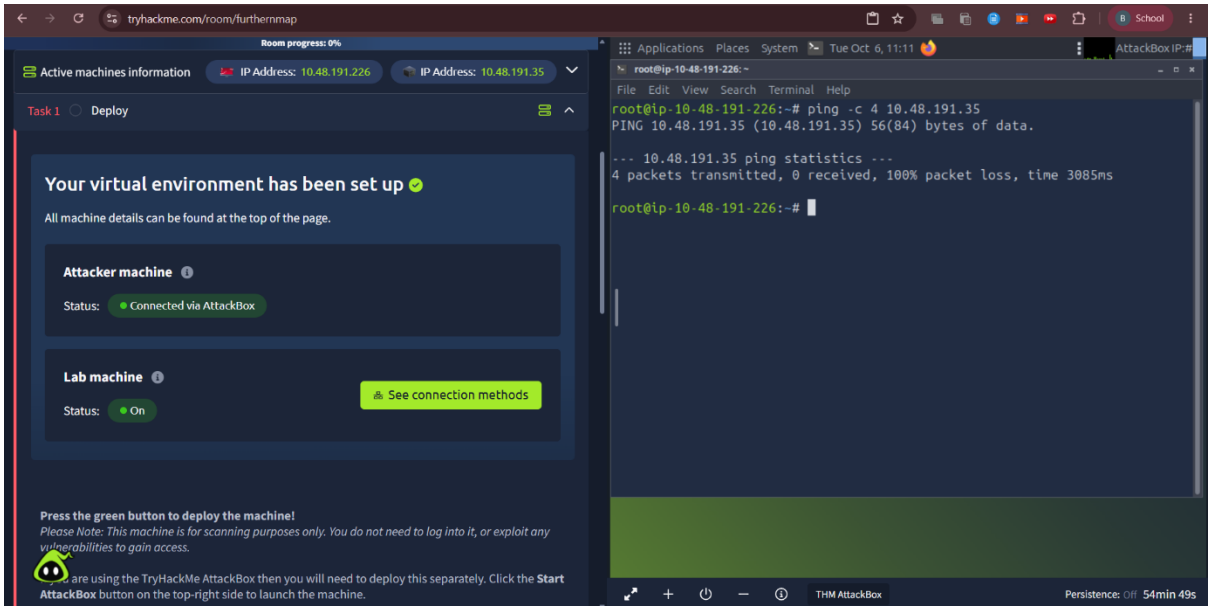
nmap <TARGET_IP>

Examine the output for host availability, scanned ports, port states and the services identified by Nmap. Record the results for comparison with the previous discovery methods.

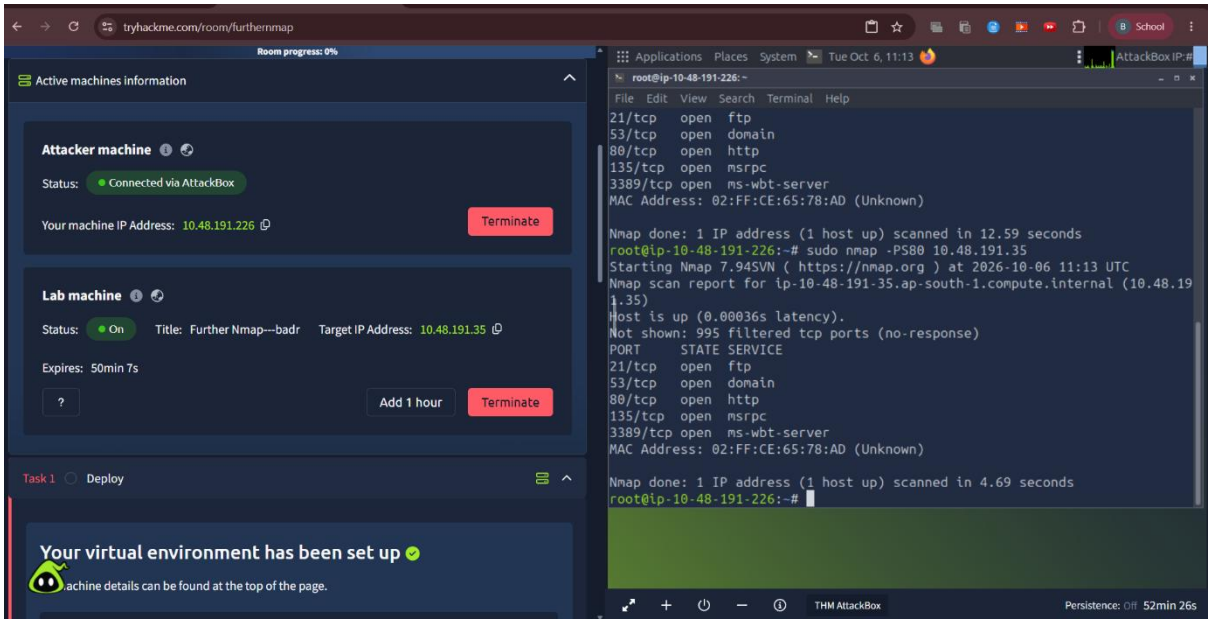
9. Compare the outputs of the different scans. Identify which methods detected the target and note any differences. These differences may occur because of network boundaries, filtering rules, disabled services or firewall configurations.

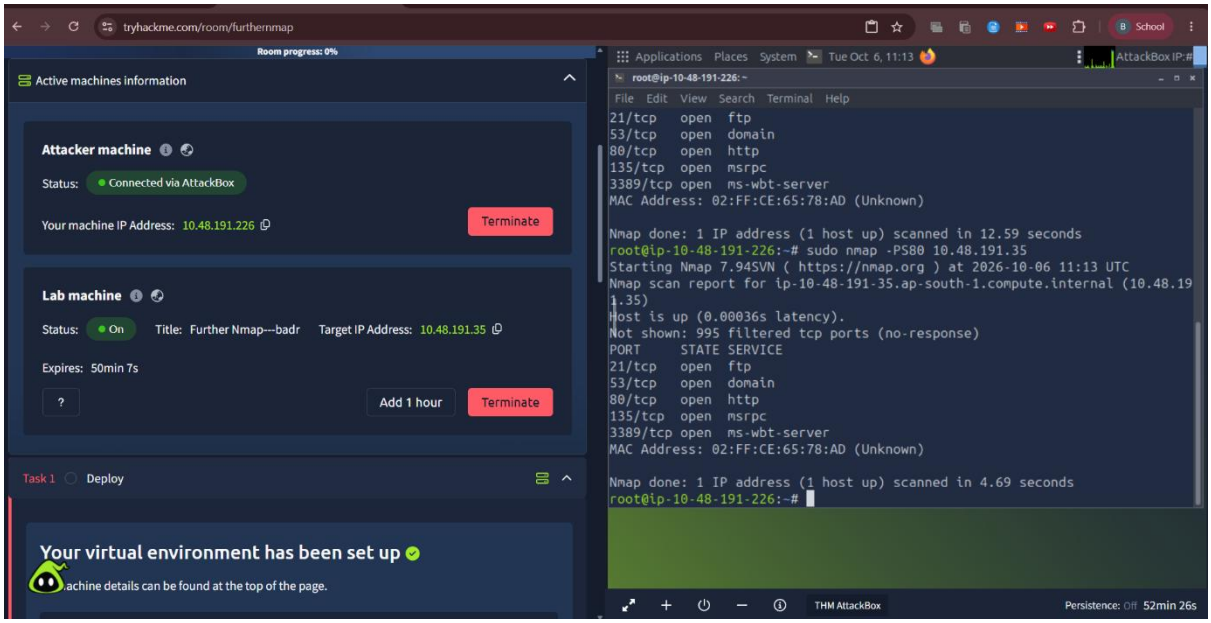
10. After completing the experiment, stop the lab machine if required and record the observed results in the laboratory record.



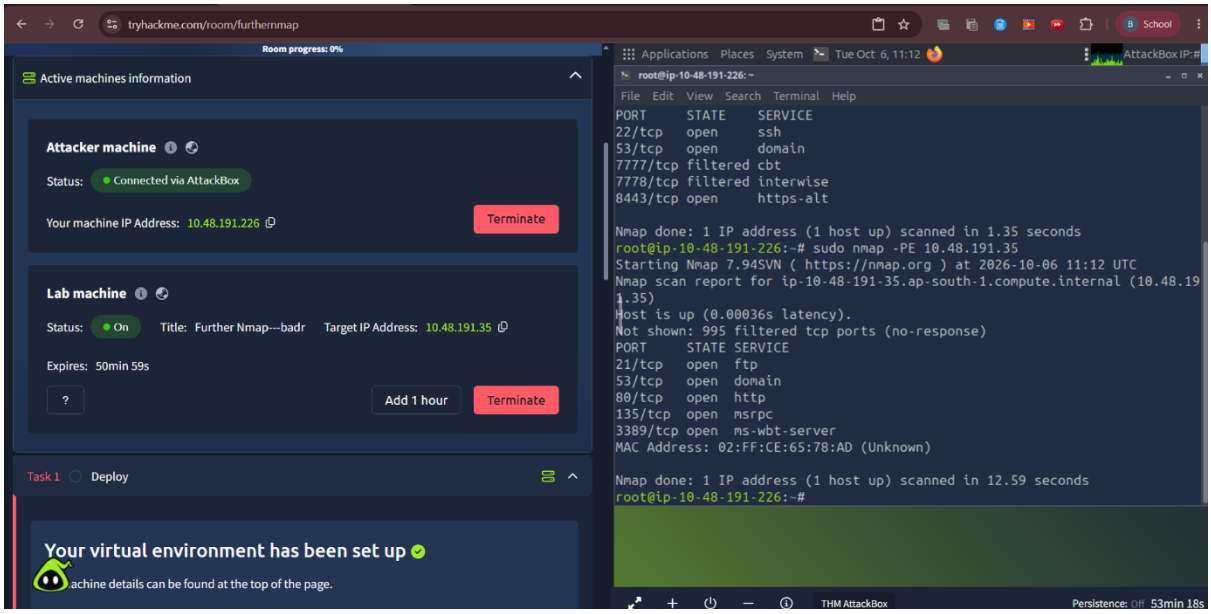


016-Akshaya K





016-Akshaya K




```
root@ip-10-48-191-226:~# sudo nmap -PU53 10.48.191.35
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-06 11:15 UTC
Nmap scan report for ip-10-48-191-35.ap-south-1.compute.internal (10.48.191.35)
Host is up (0.00034s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT     STATE SERVICE
21/tcp   open  ftp
53/tcp   open  domain
80/tcp   open  http
135/tcp  open  msrpc
3389/tcp open  ms-wbt-server
MAC Address: 02:FF:CE:65:78:AD (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 4.79 seconds
root@ip-10-48-191-226:~#
```

```
root@ip-10-48-191-226:~# nmap 10.48.191.226
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-06 11:15 UTC
Nmap scan report for ip-10-48-191-226.ap-south-1.compute.internal (10.48.191.226)
Host is up (0.000013s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE    SERVICE
22/tcp   open     ssh
53/tcp   open     domain
7777/tcp filtered cbt
7778/tcp filtered interwise
8443/tcp open     https-alt

Nmap done: 1 IP address (1 host up) scanned in 1.32 seconds
root@ip-10-48-191-226:~#
```

Result

Thus, Nmap was used on the TryHackMe platform to study host discovery through ARP, ICMP, TCP and UDP scanning techniques. The scan outputs were examined and compared to understand how different probing methods identify reachable hosts and provide information about network availability and port states.

016-Akshaya K